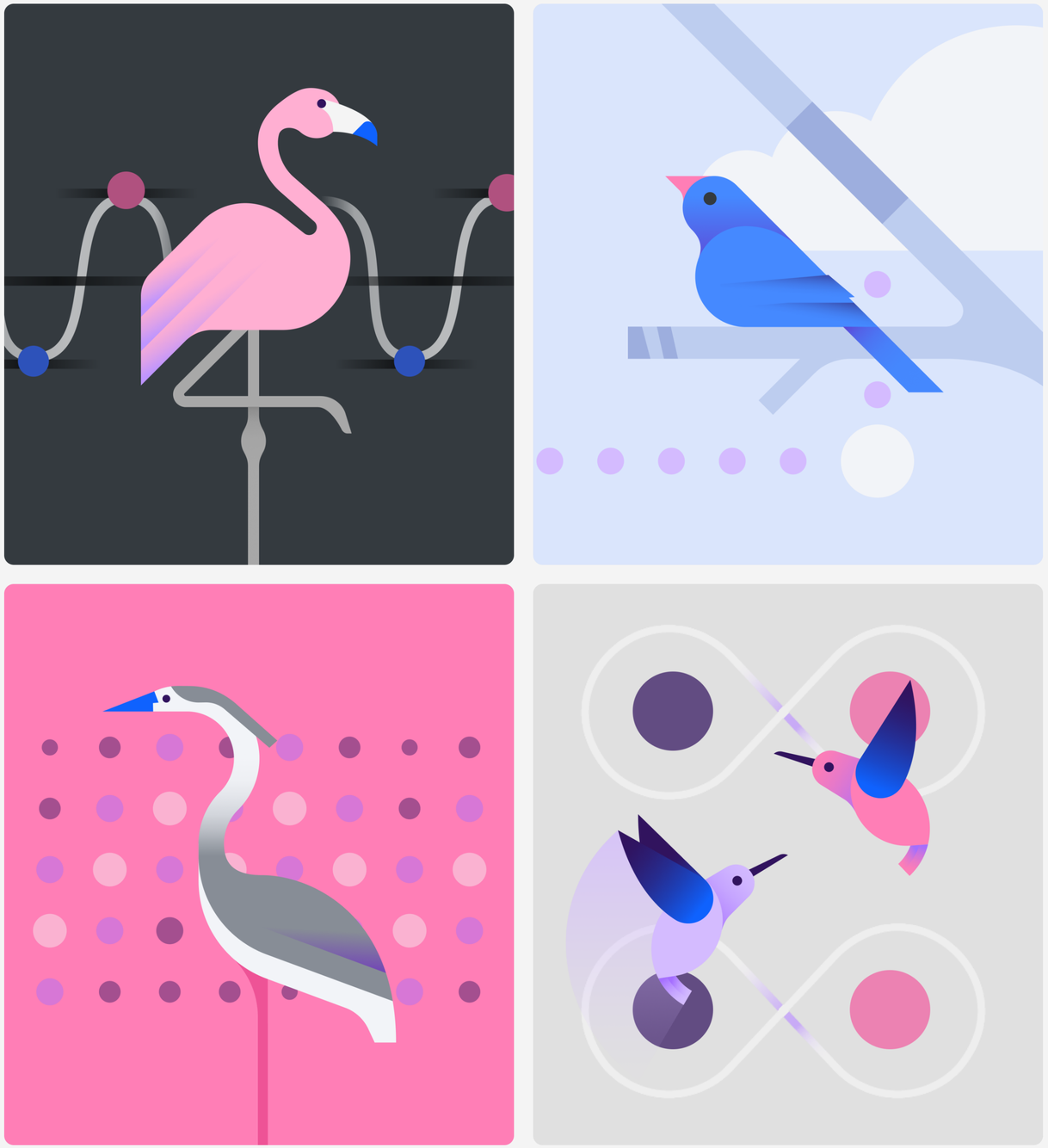
Qiskit Fall Fest 2026

Herding quantum cats



Afonso Azenha

Beginner/Intermediate

Hackathon Challenge

Qiskit Fall Fest @
University of Stuttgart

IBM

Ten years of qubits on the cloud



Falcon
27 qubits



Hummingbird
65 qubits



Eagle
127 qubits



Heron r1
133 qubits



Nighthawk r1
120 qubits

2016

IBM puts 5 qubits online: anyone with a browser can program a quantum computer.

2017

ibmqx2 “Yorktown”: about 2% two-qubit gate error and 6% readout error.

2025

IBM entangles 120 qubits in one cat state.

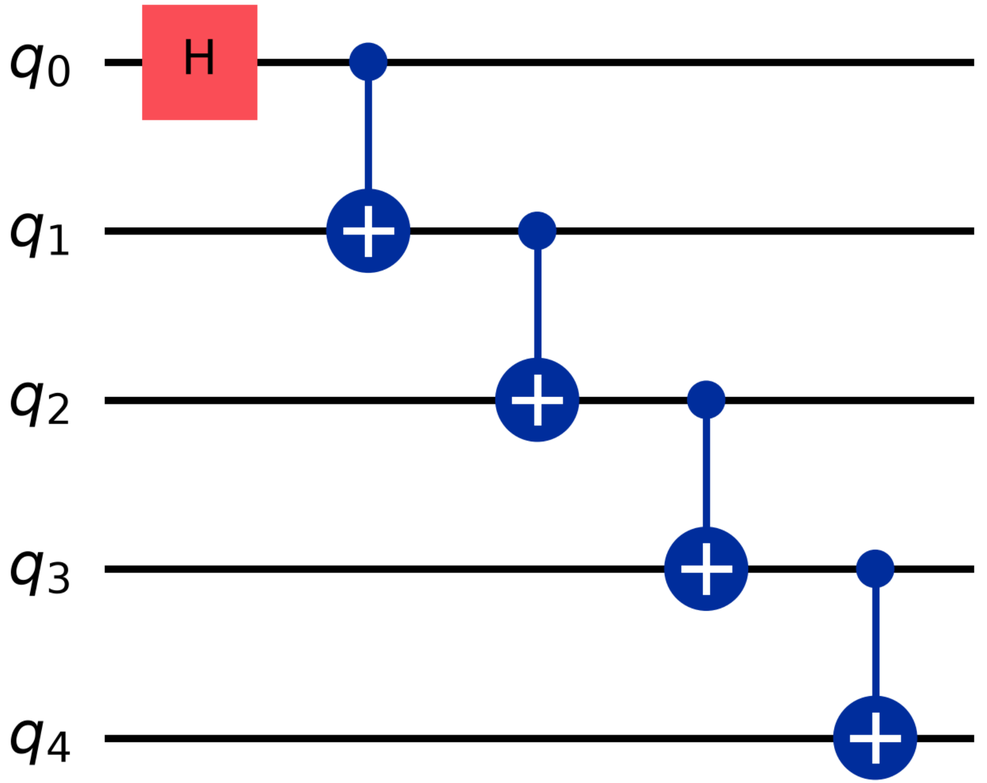
2026

ibm_kingston, a 156-qubit Heron, is open to every free account.

A cat state: all 0 and all 1 at once

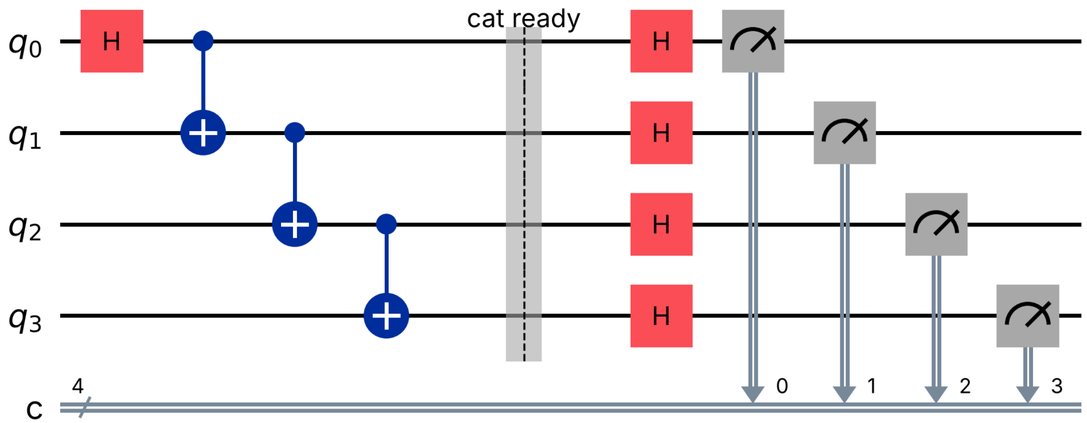
$$|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$$

- N qubits that are all 0 and all 1 **at the same time**, like Schrödinger's cat.
- **Implementation**: a Hadamard gate and a chain of CNOTs.
- **Hard on a real chip**: one error on any of the N qubits spoils it. That makes it a standard test for quantum computers.



How do we know it’s a cat?

Two measurements, one number



X basis: a Hadamard on every qubit, then measure. Z basis: measure right away.

Z basis → P

Do all qubits agree: all 0 or all 1?

X basis → X

Are the two halves in superposition, or is it just a coin flip?

Fidelity

F = P + (X – 1) / 2. A perfect cat scores 1.

Proof of entanglement

A coin flip scores 0.5. **F clearly above 0.5 proves all N qubits are entangled.**

On a **simulator** (modelling **`FakeKingston`**), Qiskit's default settings produce a 12-qubit cat that isn't even certified.

0.44

It sits on a qubit that reads out wrong 17% of the time. **On the real chip the noisy qubits change daily**: finding them is your first task.

17%



Fall Fest simulator with ibm_kingston's calibration data

Two ways to fight noise:
(Actually, there’s more…)

Error suppression

Change the circuit (**runtime**) so that less goes wrong:

- Pick good qubits (**layout**).
- Keep the circuit short.
- Protect waiting qubits with **dynamical decoupling**.
- Randomize errors so they behave predictably (**twirling**).

Error mitigation

Correct the result afterwards (**post-processing**):

- Undo readout errors (**TREX**)
- Measure at deliberately amplified noise, then extrapolate back to zero noise (**ZNE**).
- Model device noise explicitly and attempt to invert it (**PEC**), etc.

From 0.44 to about 0.98, step by step

Increase in F… Happy! ☺

0.72	0.74	0.93	~0.98
Pick good qubits (**layout**).	Grow the cat from the middle: less waiting.	**Correct readout errors**.	**Zero-noise extrapolation**.

Nine missions, from guided to open

Difficulty picks up as you move along…

Guided

0 Your first cat
1 A 2017 chip vs. a 2026 chip
2 **Layout**: where should the cat live?
3 **Depth**: grow it faster
4 **Error suppression**
5 **Error mitigation**

Open

6 Your **cleanest** 12-qubit cat
7 **The real chip**: ibm_kingston
8 **The biggest cat**

Two leaderboards

Cleanest cat

Exactly 12 qubits, as close to F = 1 as possible: on the **simulator** and on the **real chip**.

Everyone gets the same conditions: we re-run your recipe ourselves, and on hardware our baseline runs in the same job as your cat.

Biggest cat

The largest cat you can prove is entangled. Counts fully on the **real chip**, **half on the simulator**.

Judges also read your short answers and your evidence of what helped.

Simulator all afternoon, real chip at the end

10

minutes of QPU time per person on the free plan. A final job needs well under one; the queue is the bottleneck.

18:45

Last call for your final hardware job. Test your account early with a tiny job. (You can still submit after 18:45, but beware of the queues!)

20:00

Hackathon ends! One person per team posts the notebook, the submission file and a 5-minute pitch deck in Discord.



Awards tomorrow at 9:45

Go herd some cats.

Github repository:
https://github.com/kaiuki2000/herding-cats

Open **herding_cats_challenge.ipynb** and run the setup cell. Good luck, have fun! 😸 😹 😼

